

Jahn–Teller Effect in Isomerization of Cubane

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Cubane has been experimentally reported to isomerize to cuneane in the presence of Ag(I) Lewis acid catalysts. A substituted cubane yields product isomers, and their formation ratio depends on the introduced substituents. The reason for such a substituent effect has been veiled. In the present study, we determine a reaction path of the isomerization using density functional theory calculations and analyze the obtained reaction mode. Cubane becomes cationic by charge transfer to Ag(I), resulting in the Jahn–Teller effect. The cationic cubane can be thermally transformed between some stable equivalent structures on the Jahn–Teller potential energy surface (JTPES), and such a transformation (pseudo-rotation) gives rise to various patterns of C–C cleavages. Further coupling of a skeleton-twisting mode to the C–C cleavages results in the formation of cuneane. This result implies that introducing substituents asymmetrically warps the JTPES and makes the stable structures energetically non-equivalent, consequently making a difference in the formation ratio of isomers.

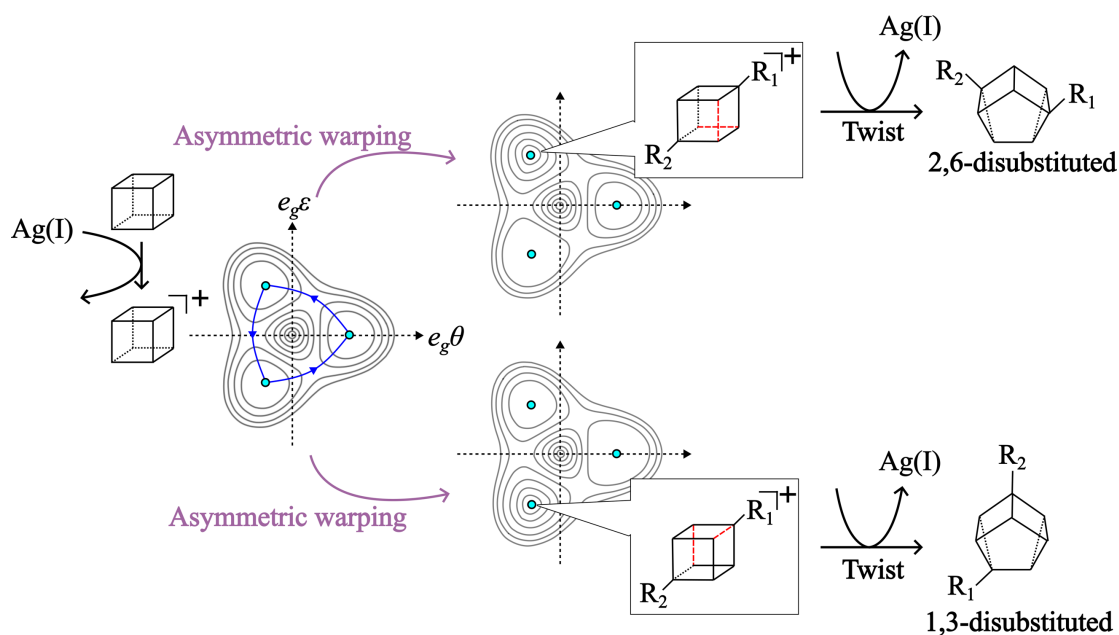


Fig. 1. The schematic diagram of isomerization reaction paths.

1) H. Takebe and S. Matsubara, Eur. J. Org. Chem. **27**, e202300891 (2024).